We know the admittance $\underline{Y} = \frac{1}{6} - \frac{1}{2\sqrt{3}}j$[....1.]. This is obtained by connecting in parallel a resistance with ...2...The susceptance, representing the ......3....part of the admittance, can be computed with the formula......4............ If the admittance is crossed by a current $\underline{I} = 12e^{-j\pi}$[5...], of frequency f=10[...6..],find the value of the voltage drop U on this admittance 7. Write the sinusoidal expression of the voltage, u(t) 8, and compute its value when t=1[ms] 9. Write the value 10 and the measurement 11 unit for the element connected in parallel with the resistance. State the measurement units for the following quatity: μ_0 12.

Laural.Darabant@gmail.com

Ora: 16:26

Laural.Darabant@gmail.com

Ora: 16:26

1. Siemens, S – 0,8
2. a coil- 2
3. Imaginary - 2
4. B=1/ωL 2
5. A, Ampere -0,8
6. Hz (Hertz) -0,8
7. $\underline{U} = \frac{I}{\underline{Y}} = \frac{12e^{-j\pi}}{\frac{1}{6}-\frac{1}{2\sqrt{3}}j} = \frac{12e^{-j\pi}}{\frac{1}{3}e^{-j\frac{\pi}{3}}} = 36e^{j(-\pi+\frac{\pi}{3})} = 36e^{-j\frac{2\pi}{3}}[V]$ -3
8. $u(t) = 36\sqrt{2}\sin\left(\omega t - \frac{2\pi}{3}\right)$ 2
9. $u(1ms) = 36\sqrt{2}\sin\left(2\pi \cdot 10 \cdot 1 \cdot 10^{-3} + \frac{\pi}{4}\right) \cong \sin(\pi \cdot 0{,}45) \cong 8{,}35[V]$ - 3
10. and 11 . $\frac{1}{\omega L} = B = \frac{1}{2\sqrt{3}} \rightarrow L = \frac{2\sqrt{3}}{\omega} = \frac{2\sqrt{3}}{2\pi \cdot 10} = 55{,}06[mH]$ 2 and 0,8

12. H/m - 0,8